

# Clinical Risk Assessment Report: Patient 42

## 1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 13.6% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.61x the diagnostic threshold

### Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Tumor Sphericity (CT) (SHAPE\_Sphericity(onlyFor3DROI)).CT), which reduced the patient's absolute risk by 1.6%. Biologically, this feature quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

## 2. Key Pathological & Imaging Drivers (Elevating Risk)

**Small Zone Emphasis (PET) (GLZLM\_SZE.PET)** **Value: 1.36 [Top 8%] (+1.5%)**  
*Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.*

## 3. Protective / Mitigating Factors (Reducing Risk)

**Tumor Sphericity (CT) (SHAPE\_Sphericity(onlyFor3DROI)).CT)** **Value: 1.65 [Top 3%] (-1.6%)**  
*Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.*

**Patient Age (Age)** **Value: 65.0 [Avg (54th %ile)] (-1.5%)**  
*Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.*

**75th Percentile Tumor Density (CONVENTIONAL\_HUQ3)** **Value: 0.33 [Avg (52th %ile)] (-1.0%)**  
*Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.*

**Short Run High Gray-Level Emphasis (CT) (GLRLM\_SRHGE.CT)** **Value: 0.32 [Avg (54th %ile)] (-0.9%)**  
*Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.*

**Squamous Cell Carcinoma Antigen (SCC\_Ag)** **Value: 1.7 [Top 3%] (-0.7%)**  
*Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.*

**Small Zone High Gray-Level Emphasis (PET) (GLZLM\_SZHGE.PET)** **Value: 0.28 [Top 18%] (-0.1%)**  
*Indicates multiple small areas of intense density or metabolism, a marker of erratic and aggressive cellular proliferation.*

**Long Run High Gray-Level Emphasis (PET) (GLRLM\_LRHGE.PET)** **Value: 0.14 [Avg (75th %ile)] (-0.1%)**  
*Points to continuous, elongated bands of high-density or highly active tumor tissue.*

Machine Learning Feature Attribution (SHAP Decision Path)

